



Fig. 2: Parts in a basket. Every cleaning scoreboard needs to be tailored to the task at hand. For example, cleaning large metal parts with simple shapes is an easy application that might be handled nicely by an aqueous or hydrocarbon cleaner. Conversely, as parts get smaller or more complex in their shapes, vapour degreasers have a performance edge

CPPC is a number, ranging from hundreds of a rupee to hundreds of them. This score summarises all of the direct cleaning costs, environmental inputs, labour costs and waste disposal costs into one simple question: how much does it cost to clean a single part? This cost becomes the tool that allows the savvy engineer to compare totally different cleaning technologies fairly.

So here's the crucial teaching of this report: stop caring about prices. It is not the 'rupee-per-litre' that really matters; it is the 'total-rupee-per-part-cleaned.' That is where profits will be made or lost.

For the purpose of this discussion, the cost-per-part-cleaned rating is the most practical. However, customers may prefer cleaning-cost-per-day or cost-per-shift as an equally effective metric of efficiency. In general, boards-per-hour is optimal for most companies because most of the other operational expenses are either recorded as hourly numbers or can be easily converted into hourly costs.

Take a practice swing

When buying a new baseball bat, tennis racket or golf club, people select the one that fits their strength, size and swing. Similarly, cleaning machines come in all different sizes and with

varying features. It is important to select the cleaning system that best fits the company's cleaning application. This requires a four-step process.

Size. The first parameter is size. If a system is too small, there will be a backlog. If the system is too large, it will be sitting idle, which is money wasted. The 'fit' of a cleaning system is judged by its throughput,

measured in boards-per-hour.

As an example—a batch aqueous machine that cleans 20 boards in 40 minutes has an average throughput of one board every two minutes, which means about 30 boards per hour. A batch vapour degreaser that cleans only five boards at a time but cleans them in five minutes, will have a throughput of 60 boards per hour.

Throughput affects operating expenses dramatically, especially as cleaning systems reach their operational limits. For example, a small, latest vapour degreaser should have solvent losses that do not exceed about 20 grams per hour. In standby mode, this drops to about 4 grams per hour. To compute total cleaning costs, engineers will need to estimate how many hours each day the system will be cleaning and how many hours the system will be idle.

Many factors affect throughput. For example, systems that need to pause while warming up will have longer cycles. Complex loading and unloading processes also delay the cycle. Drying times may delay cleaning cycles. Water-based machines often cycle quickly when cleaning large, simple shapes; vapour systems are faster when cleaning tight spaces or components with many voids.

Performance. Engineers should

explicitly define the level of cleanliness required and then ensure each system can achieve that requirement. Simple visual inspection may be sufficient in many instances; surface insulation testing may be essential for more demanding applications. Whatever the standard is, define it before the testing stage.

Convenience. Engineers must consider where the cleaning will occur. For example, benchtop machines are slow, small and cheap, so it's easy to have them conveniently scattered around the plant. Centralised high-volume systems are larger, more capable, more efficient and more expensive, but technicians waste time taking PCBs over to the centralised cleaning system. Convenience is complex.

Test and verify. Lastly, it is essential to conduct real-world tests of the systems. Prepare a batch of products with typical contamination loads. Have each system manufacturer run them through their cleaning systems. Double-check to ensure that cleaning did not degrade components or leave

Checklist of cleaning costs

Operating costs

1. Labour: Operator cost per hour (fully-loaded labour rate)
2. Electricity
3. Water, solvents or saponifiers
4. Consumables (filters, etc)
5. Water-treatment system consumables
6. Waste disposal

Indirect operating costs

1. Labour: System maintenance cost per hour
2. Floor space cost per square metre
3. Training of operators
4. Supervisory staff
5. Lighting, ventilation and other services

One-time capital costs

1. Cost of buying the cleaning system
2. Freight, duties, customs fees and insurance to get it delivered
3. Site engineering and architectural planning costs
4. On-site construction
5. Electrical changes
6. Water/plumbing changes
7. Ventilation changes
8. System setup and configuration (delivery)
9. Cost of capital